

Project Design Phase-II

Technology Stack (Architecture & Stack)

Parameter	Details
Date	12 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	4 Marks

Technical Architecture

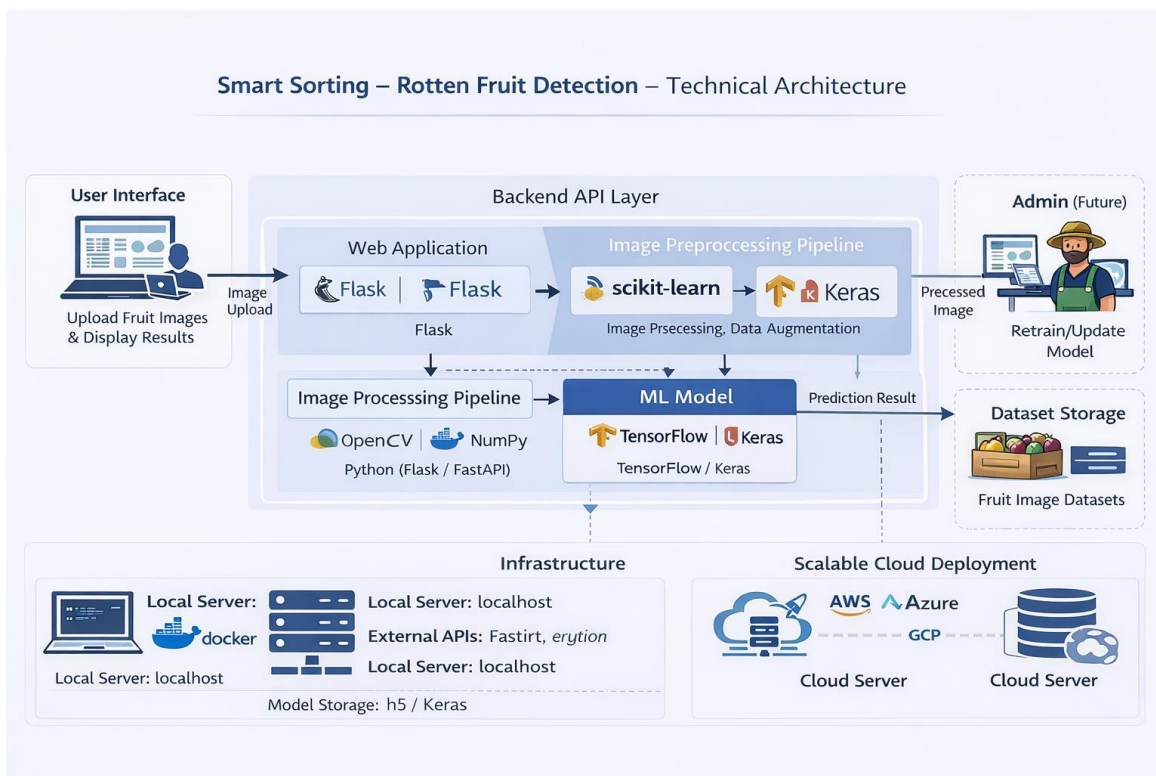


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Image upload interface and prediction result display	HTML, CSS, JavaScript
2	Application Logic	Handles image upload requests, validation, and prediction routing	Python (Flask)

3	Image Preprocessing Pipeline	Image resizing, normalization, and augmentation	OpenCV, NumPy
4	Machine Learning Model	Fruit classification model (Fresh vs Rotten)	TensorFlow / Keras (CNN)
5	Dataset Storage	Stores fruit image datasets used for training	Local File System
6	Model Artifact Storage	Stores trained model file (.h5 / .keras)	Local File System
7	External Integration (Future)	Integration with automated fruit sorting machines	IoT / Camera Systems (Future)
8	Infrastructure	Local deployment with possible cloud deployment	Localhost, Docker (Future), AWS / Azure / GCP

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Machine learning and backend built using open-source technologies	Flask, TensorFlow, NumPy
2	Security Implementations	Secure handling of uploaded images and input validation	HTTPS, Input Validation
3	Scalable Architecture	Modular design allowing future system expansion	Layered architecture
4	Availability	System accessible through web interface for prediction	Local Web Server
5	Performance	Fast image classification with low latency prediction	Optimized ML inference